

RAI Risk Taxonomy

Technical Report 2.0

Staged classification, review-hold policy, and statistical validation
of 1,726 global L4 risks

July 21, 2026

Final policy result. Of 1,726 L4 risk cards, 1,671 (96.8134%) are provisionally assigned to the 50 L3 nodes. The 55 most uncertain cards (3.1866%) are separated into the non-taxonomic **RAI-HOLD** review bucket. No algorithmic proposal in this report constitutes human-approved ground truth.

1 Technical summary

This report documents the staged placement of 1,726 global L4 risk cards into a new hierarchy containing 50 L3 nodes. The 182 classifications inherited from the Physical AI taxonomy are treated as a locked gold set. The legacy global L1–L3 hierarchy is retained only as reference metadata and is not used as a prediction feature. Permanent L4 identifiers and source crosswalks are preserved throughout the process.

Stage 1 applies strict rule–semantic–anchor agreement and requires two expert agents to agree on the exact L3. It retains the 182 Physical locks and proposes only 37 additional placements. Stage 2 relaxes the open-set threshold and places 1,553 cards (89.9768%), leaving 173 for taxonomy review. Stage 3 tests 100% operational coverage by forcing all 173 remaining cards to their recorded hierarchy-blind top-1 L3. The final policy view then removes the 55 hard holds from the taxonomy and retains 1,671 placements plus 55 review holds.

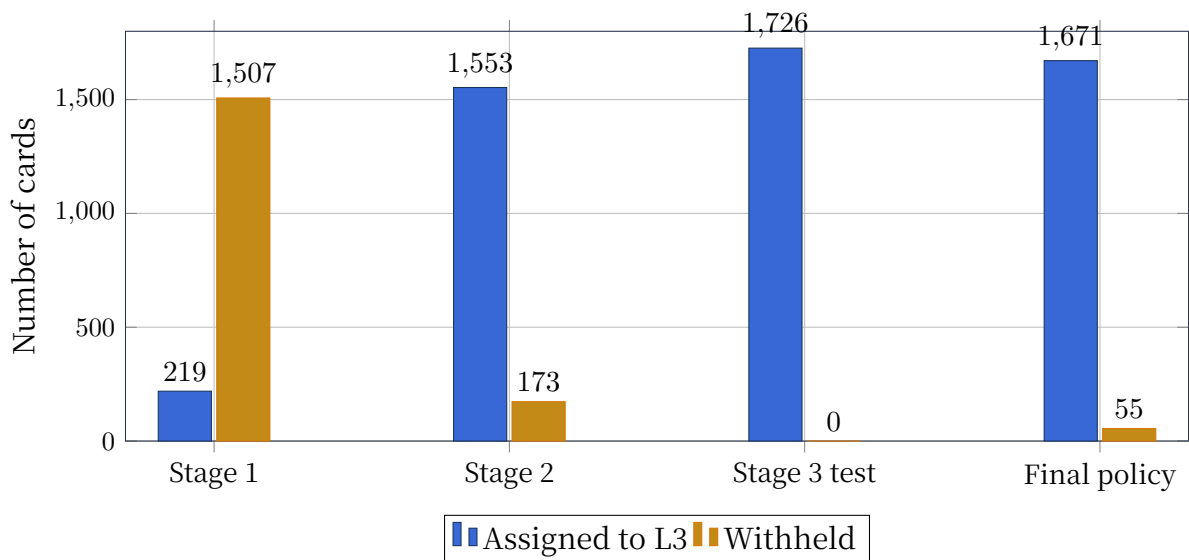


Figure 1: Assigned and withheld cards at each stage. The denominator is 1,726 throughout.

Stage 3's 100% figure measures operational path completeness, not validity. The final 96.8134% policy coverage reflects the deliberate separation of boundary conflicts. Neither figure is an accuracy or approval rate.

2 Scope, data, and identification rules

The unit of analysis is one L4 card with a permanent RAI4 identifier. The identifier range runs from 0001 through 1726. Labels are not used as identifiers because identical names may occur on different paths. The hierarchy contains one L0 root, three L1 domains, six L2 groups, and 50 L3 nodes. **RAI-HOLD** is an operational review bucket and is not an L3 node.

Final policy status	Cards	Share of all cards
Locked Physical gold set	182	10.5446%
Stage 1 expert-consensus proposal	37	2.1437%
Stage 2 algorithmic proposal	1,334	77.2885%
Stage 3 residual forced candidate	118	6.8366%
Taxonomy Decision Hold	55	3.1866%
Total	1,726	100.0000%

Table 1: Composition of the final policy view.

3 Staged classification process and decision criteria

3.1 Stage 0: canonicalization and protected sets

Before classification, the global and Physical sources are joined through the source cross-walk and assigned permanent RAI4 IDs. The Physical AI site’s 182 existing L4-to-L3 classifications form a protected gold set. Their paths cannot be changed by later algorithmic stages. The remaining 1,544 cards enter the hierarchy-blind classification pipeline.

3.2 Stage 1: strict open-set classification

For each non-Physical card, the model input contains only the L4 label, a direct mechanism definition with hierarchy restatements removed, and the evidence title. Legacy global L0–L3 values, source family, and source-ID prefixes are excluded from prediction features. A Stage 1 automatic candidate must satisfy all of the following criteria:

1. Exactly one L3 is rule-eligible.
2. The rule winner, semantic top-1, and composite top-1 are the same L3.
3. Top-1 semantic score is at least 0.55 and composite score is at least 0.54.
4. Both semantic and composite margins are at least 0.035.
5. At least two of three anchor views select the same L3 as top-1.
6. No taxonomy-gap sentinel, multi-mechanism flag, or Physical-boundary flag is present.
7. Two independent expert agents approve the same exact L3.

This gate yields 37 new expert-consensus proposals. Together with the 182 Physical locks, Stage 1 assigns 219 cards and withholds 1,507.

3.3 Stage 2: relaxed classification targeting 90% coverage

Stage 2 preserves all 219 protected placements and ranks the Stage 1 holds with the deterministic suitability score

$$S_2 = 0.55q_{sem} + 0.20q_{sm} + 0.10q_{cm} + 0.10q_{vote} + 0.05q_{rule} - p_{gap}.$$

The normalized terms are defined as

$$q_{sem} = \text{clip}((s_{sem} - 0.45)/0.25, 0, 1),$$

$$q_{sm} = \text{clip}(m_{sem}/0.05, 0, 1),$$

$$q_{cm} = \text{clip}(m_{comp}/0.05, 0, 1),$$

$$q_{vote} = v_{anchor}/3,$$

$$q_{rule} = \mathbf{1}[\text{at least one eligible L3}],$$

$$p_{gap} = 0.05 \mathbf{1}[\text{gap sentinel present}].$$

The score is a review-ordering measure, not a calibrated probability. Stage 2 first preserves 55 hard holds regardless of score. It then sorts all remaining holds by ascending S_2 , using permanent RAI4 ID as the deterministic tie breaker, and reserves the bottom 118. The final Stage 2 hold set is therefore $55 + 118 = 173$ cards (10.0232%). All other cards are assigned to their recorded hierarchy-blind top-1 L3, producing 1,553 placements (89.9768%).

3.4 Stage 3: forced matching test

Stage 3 does not retrain or rescore the model. It preserves the 1,553 non-null Stage 2 placements and assigns only the 173 Stage 2 holds to the top-1 L3 already recorded in the algorithm score artifact. Thresholds, margins, rule eligibility, and gap sentinels no longer block assignment. Every new placement is marked `forced_match_stage3`; the original hold reason, hard/quota class, score, review priority, and `human_approved=false` flag remain visible. This experiment achieves 1,726/1,726 operational paths but does not establish that the 50 L3 taxonomy is sufficient.

3.5 Final policy: separate the 55 hard holds

The final policy view reverses the forced taxonomy placement for five hard-hold families. Their policy L3 is set to null and their operational bucket is set to **RAI-HOLD**. The 118 quota holds remain forced candidates requiring human review. The forced top-1 L3 for each of the 55 holds remains available only as audit metadata.

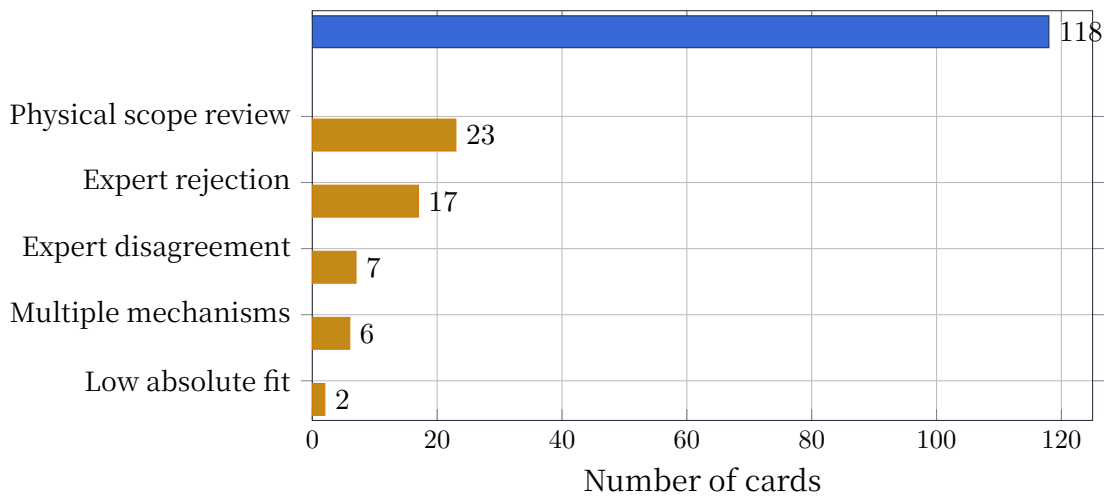


Figure 2: Reasons for the 173 Stage 2 holds. The gold 55 become **RAI-HOLD**; the blue 118 remain Stage 3 forced candidates.

The 23 Physical-scope cases are not confirmed Physical AI cards. Their text triggered terms such as robot, drone, autonomous vehicle, embodied AI, cyber-physical, or physical actuator, but they were not members of the locked Physical gold set. They therefore require a human boundary decision.

4 Final distribution statistics

4.1 L1 and L2 distribution

Among the 1,671 cards with a policy L3, General contains 1,140 cards (68.2226%), Agentic AI contains 349 (20.8857%), and Physical AI contains 182 (10.8917%). The remaining 55 cards equal 3.1866% of all 1,726 cards.

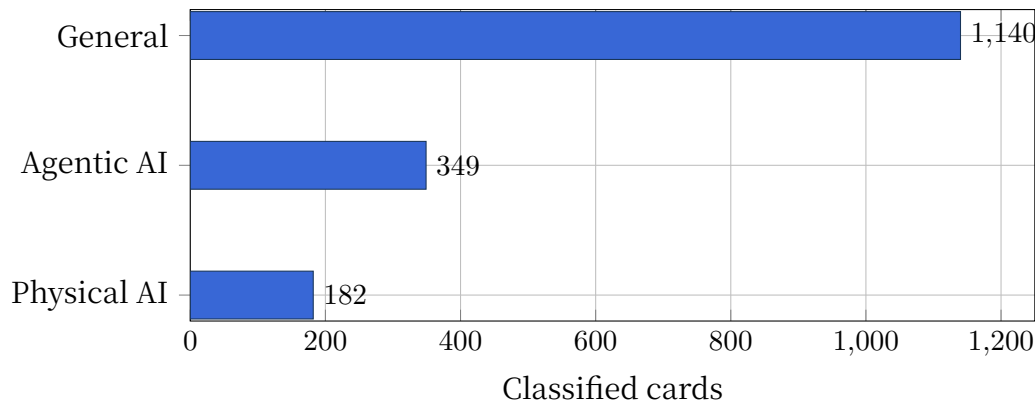


Figure 3: L1 distribution in the final policy view. The denominator is 1,671 classified cards.

L2 ID	L2 label	Cards	Share of 1,671
RAI2-G-INT	General Interaction	708	42.3698%
RAI2-G-SYS	General System	432	25.8528%
RAI2-A-SYS	Agentic System	349	20.8857%
RAI2-P-SYS	Physical System Safety	91	5.4458%
RAI2-P-INT	Physical Interaction Safety	62	3.7104%
RAI2-P-SOC	Physical Societal Safety	29	1.7355%

4.2 L3 concentration

All 50 L3 nodes contain at least one card. Anthropomorphism is the largest L3 with 242 cards, or 14.4823% of the 1,671 classified cards. The top five L3 nodes contain 700 cards (41.8911%). No individual L3 exceeds the 20% concentration warning threshold. Concentration describes distribution shape; it does not validate semantic correctness.

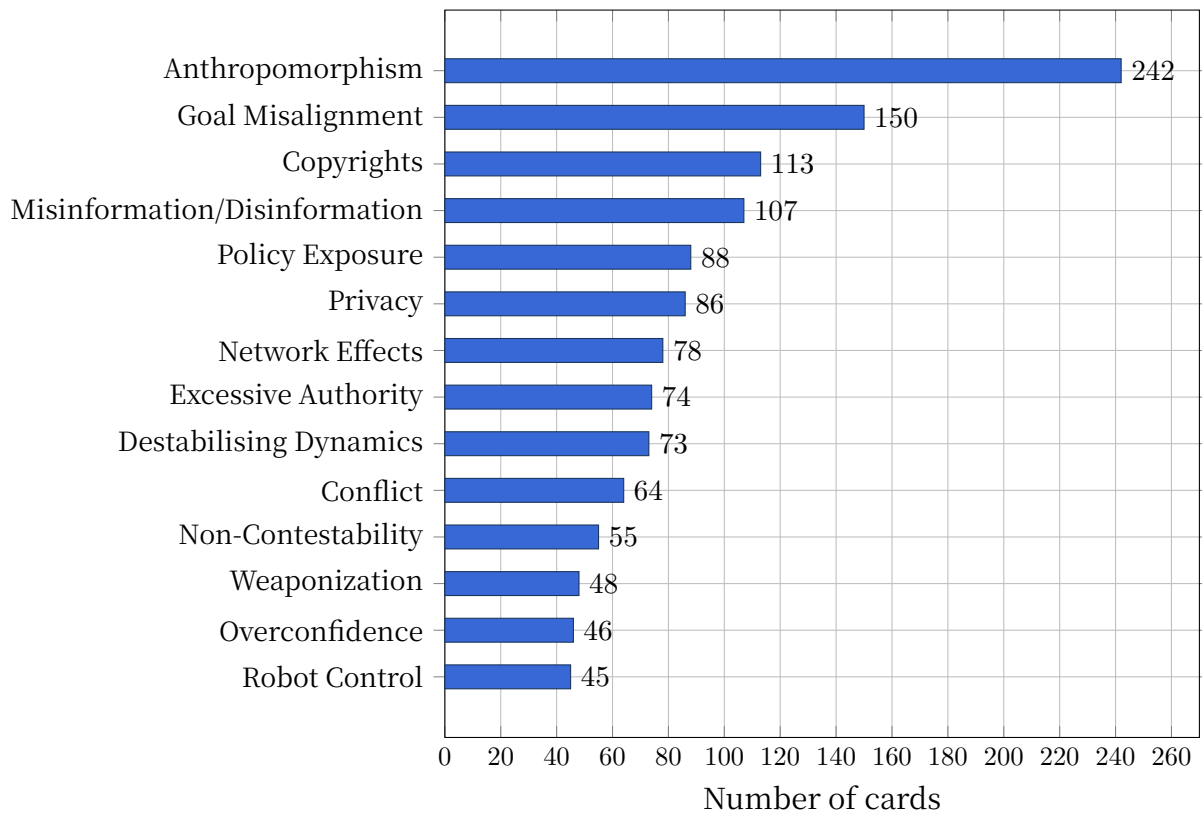


Figure 4: Top 14 L3 nodes in the final policy view. The complete 50-node distribution is available in the CSV artifact.

5 The 55 review holds and uncertainty

Hold reason	Cards	Share of holds	Share of all cards
PHYSICAL_OUTSIDE_LOCK	23	41.8182%	1.3326%
FRONTIER_EXPERT_REJECTED	17	30.9091%	0.9849%
FRONTIER_EXPERT_DISAGREEMENT	7	12.7273%	0.4056%
MULTI_MECHANISM	6	10.9091%	0.3476%
LOW_ABSOLUTE_FIT	2	3.6364%	0.1159%
Total	55	100.0000%	3.1866%

The hold bucket must not be presented as an unapproved 51st L3. The website and exports should show it as a separate review queue and retain the original hold reason, forced-candidate L3, semantic and composite scores, anchor votes, and critical review priority. The 24 expert rejection/disagreement cases test L3 definition fit; the six multi-mechanism cases test whether primary/secondary or multi-label structure is needed; and the 23 Physical-boundary cases require comparison against the locked Physical taxonomy.

6 Validation, limitations, and robustness checks

The 182 Physical placements and 37 Stage 1 expert-consensus placements remain unchanged in all later stages. The Stage 2 and Stage 3 builders reproduce byte-identical placement hashes on repeated runs. The Stage 2 validator reports 13 PASS, 0 FAIL, and 3 WARN; the Stage 3 validator reports 14 PASS, 0 FAIL, and 4 WARN. All 16 repository regression tests, including the final hold-policy tests, pass.

Material limitations are:

- 1,214 of the 1,334 new Stage 2 placements lack Stage 1 rule eligibility and rely primarily on semantic top-1 proximity.
- 247 new Stage 2 placements override a taxonomy-gap sentinel.
- The remaining 118 Stage 3 forced candidates are neither calibrated probabilities nor human approvals.
- The Physical-boundary sentinel uses a limited keyword set and can produce both false positives and false negatives.
- Use of all 50 L3 nodes and 96.8134% card coverage do not prove taxonomy sufficiency.

7 Recommended review sequence and open questions

1. Review all 55 RAI-HOLD cards, prioritizing the 23 Physical-boundary cases and 24 expert rejection/disagreement cases.
2. Review the 118 Stage 3 quota candidates in ascending S_2 order.
3. Audit stratified L3 samples from the 247 gap overrides and 1,214 rule-free placements.
4. Use the evidence to select a remedy: expand an L3, add an L3, or permit primary and secondary labels.
5. Publish only human-approved migrations. Preserve decision IDs and history in the successor release.

Open decisions include how many Physical-boundary cards belong in the Physical taxonomy, whether multi-mechanism cards may receive a secondary L3, what human review error tolerance is acceptable for the 118 forced candidates, and what minimum evidence should be required for proposing a new L3.

Reproducible artifacts

- data/experiments/stage2-v1/stage2_placements.json
- data/experiments/stage2-v1/stage2_hold_queue.json
- data/experiments/stage3-v1/stage3_placements.json
- data/experiments/stage3-review-hold-v1/placements.json
- data/experiments/stage3-review-hold-v1/review_hold.json
- reports/statistics/stage2_unclassified_statistics.json
- reports/statistics/stage3_statistics.json
- scripts/build_stage2_classification.py
- scripts/build_stage3_forced_matching.py
- scripts/build_stage3_review_hold.py

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Language edition: English

Policy data view: stage3-review-hold-v1

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